

Exercise 1: Four-velocity

- a) The general definition of the four-velocity is

$$u^\mu = \begin{pmatrix} \gamma c \\ \gamma u_x \\ \gamma u_y \\ \gamma u_z \end{pmatrix}. \quad (1)$$

In the particle's restframe $\gamma = 1$, so that

$$u^\mu = \begin{pmatrix} c \\ 0 \\ 0 \\ 0 \end{pmatrix}. \quad (2)$$

- b) After a Lorentz transformation from $\mathbf{K}$ to $\mathbf{K}'$, the four-velocity reads

$$u^{\mu'} = \begin{pmatrix} \gamma c \\ -\gamma \beta c \\ 0 \\ 0 \end{pmatrix}. \quad (3)$$

where the x axis was defined as being parallel to $\vec{v}$ without loss of generality. In the previous equation, we have

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}. \quad (4)$$

- c) The four-velocity of the second particle in the frame of reference $\mathbf{K}$ is given by

$$u^\mu = \begin{pmatrix} \tilde{\gamma} c \\ \tilde{\gamma} u \\ 0 \\ 0 \end{pmatrix}. \quad (5)$$

But here,

$$\tilde{\gamma} = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}} \quad (6)$$

since we are computing the four-velocity of the particle with velocity $\vec{u}$. The Minkowski-norm of this vector is

$$\begin{aligned}
u_\mu u^\mu &= (\tilde{\gamma}c, \tilde{\gamma}u, 0, 0) \begin{pmatrix} \tilde{\gamma}c \\ -\tilde{\gamma}u \\ 0 \\ 0 \end{pmatrix} \\
&= \tilde{\gamma}^2(c^2 - u^2) = \tilde{\gamma}^2 c^2 (1 - \frac{u^2}{c^2}) = c^2,
\end{aligned}$$

where we used $\tilde{\gamma}$, as we are calculating in the frame $\mathbf{K}$. To obtain the norm in the reference frame $\mathbf{K}'$, we first have to use the same Lorentz transformation as in b). The four-velocity in $\mathbf{K}'$ is then given by

$$u^{\mu'} = \begin{pmatrix} \gamma(\tilde{\gamma}c - \beta\tilde{\gamma}u) \\ \gamma(\tilde{\gamma}u - \beta\tilde{\gamma}c) \\ 0 \\ 0 \end{pmatrix}. \quad (7)$$

Its norm is given by

$$\begin{aligned}
u_\mu u^\mu &= \gamma^2 [(\tilde{\gamma}c - \beta\tilde{\gamma}u)^2 - (\tilde{\gamma}u - \beta\tilde{\gamma}c)^2] \\
&= \gamma^2 [\tilde{\gamma}^2(1 - \beta^2)(c^2 - u^2)] = c^2.
\end{aligned}$$

Exercise 2: Minkowski Diagrams

- a) Let us apply the Lorentz transformations to the coordinate system $\mathbf{K}'$. For the axis ct' we find (note that $x' = 0$):

$$\begin{aligned}
0 = x' &= \gamma(x - ut) \Rightarrow x = ut = \frac{u}{c}ct \Rightarrow \frac{x}{ct} = \frac{u}{c} = \beta \\
\Rightarrow \tan \alpha_1 &= \frac{x}{ct} = \beta \Rightarrow \alpha_1 = \arctan \beta,
\end{aligned} \quad (8)$$

where α_1 is the angle between the ct and the ct' axis. Similarly for x' we find (note that $ct' = 0$):

$$\begin{aligned}
0 = ct' &= \gamma(ct - \beta x) \Rightarrow \frac{ct}{x} = \frac{u}{c} = \beta \\
\Rightarrow \tan \alpha_2 &= \frac{ct}{x} = \beta \Rightarrow \alpha_2 = \arctan \beta.
\end{aligned} \quad (9)$$

where α_2 is the angle between the x and the x' axis. So we have $\alpha = \alpha_1 = \alpha_2 = \arctan \beta$. For $\beta \in [0, 1[$ the arctan of β is between 0 and $\pi/4$ (see figure 2.1).

- b) The components of p' are

$$p' = \begin{pmatrix} \gamma ct - \gamma \beta x \\ \gamma x - \gamma \beta ct \end{pmatrix} = \gamma \begin{pmatrix} ct - \beta x \\ x - \beta ct \end{pmatrix}. \quad (10)$$

The 4-length of p' squared is therefore

$$\begin{aligned}
p'^2 &= \gamma^2 ((ct - \beta x)^2 - (x - \beta ct)^2) \\
&= \gamma^2 (c^2 t^2 + \beta^2 x^2 - 2\beta ct x - x^2 - \beta^2 c^2 t^2 + 2\beta ct x) \\
&= c^2 t^2 \gamma^2 (1 - \beta^2) - x^2 \gamma^2 (1 - \beta^2) \\
&= c^2 t^2 - x^2 = p^2,
\end{aligned} \quad (11)$$

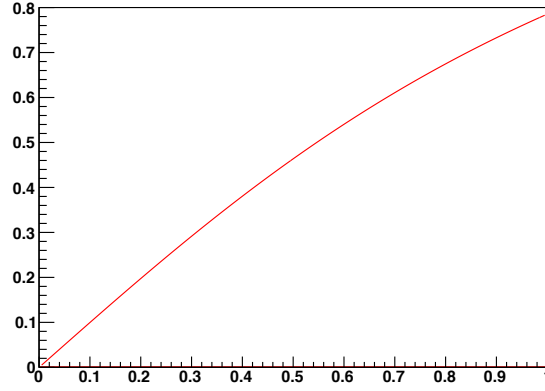


Abbildung 2.1: $\arctan x$ für $x \in [0, 1[$.

where we used that

$$\gamma^2(1 - \beta^2) = \frac{1}{1 - \beta^2}(1 - \beta^2) = 1. \quad (12)$$

c) For the coordinates of event P in system $\mathbf{K}'$ we have

$$\begin{aligned} ct'_0 &= \gamma(ct_0 - \beta x_0), \\ x'_0 &= \gamma(x_0 - \beta ct_0). \end{aligned} \quad (13)$$

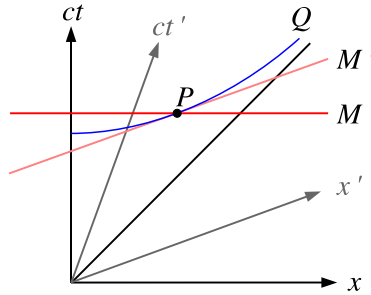


Abbildung 2.2: Events in the Minkowski diagram.

The set M of all events which occur simultaneously with P in $\mathbf{K}$ are located on a line parallel to the x axis, intersecting P . The set M' of events which occur simultaneously with P in $\mathbf{K}'$ are on a line parallel to the x' axis, intersecting the point P (see Figure 2.2).

d) The set Q of events having the same 4-distance from the origin are found on the hyperbola

$$p^2 = c^2 t^2 - x^2 \Rightarrow ct = \sqrt{p^2 + x^2}, \quad (14)$$

which you can see in Figure 2.2).

Exercise 3: Causality

- a) Using the Lorentz transformation it is possible to translate the coordinates (x_1, t_1) and (x_2, t_2) from reference frame $\mathbf{K}$ to $\mathbf{K}'$:

$$\begin{aligned}x'_1 &= \gamma (x_1 - v t_1) & t'_1 &= \gamma \left(t_1 - \frac{v x_1}{c^2} \right) \\x'_2 &= \gamma (x_2 - v t_2) & t'_2 &= \gamma \left(t_2 - \frac{v x_2}{c^2} \right)\end{aligned}$$

This can be used to show the equality

$$t'_2 - t'_1 = \gamma \left((t_2 - t_1) - \frac{v (x_2 - x_1)}{c^2} \right) \quad (15)$$

$$= \gamma \left(\Delta t - \frac{v \Delta x}{c^2} \right). \quad (16)$$

where $\Delta t = t_2 - t_1$ and $\Delta x = x_2 - x_1$.

- b) Two events happening at the same time in reference frame $\mathbf{K}'$ implies $t'_2 = t'_1$ which is equal to

$$\Delta t - \frac{v \Delta x}{c^2} = 0 \quad (17)$$

$$\Rightarrow \Delta x = \frac{c^2 \Delta t}{v}. \quad (18)$$

Since $v < c$ the inequality $\Delta x > c \Delta t$ holds true.

- c) It is required to show $\Delta x < c \Delta t \Rightarrow t'_2 > t'_1$.

$$t'_2 = \Delta t' + t'_1 \quad (19)$$

$$t'_2 = \underbrace{\gamma \left(\Delta t - \frac{v}{c^2} \Delta x \right)}_{\text{to show: } > 0} + t'_1 \quad (20)$$

The Lorentz factor γ is greater than 1 by definition. $\Delta t > \Delta x/c$ implies for the bracket

$$\Delta t - \frac{v}{c^2} \Delta x > \frac{\Delta x}{c} - \frac{v}{c^2} \Delta x \quad (21)$$

$$= \underbrace{\frac{\Delta x}{c}}_{>0} \underbrace{1/c}_{>0} \underbrace{\left(1 - \frac{v}{c} \right)}_{>0, \text{ since } v < c} . \quad \square \quad (22)$$

- d) $\Delta x = c' \Delta t > c \Delta t$ implies

$$\gamma \left(\Delta t - \frac{v \Delta x}{c^2} \right) = \gamma \Delta t \left(1 - \frac{v c'}{c} \right) = t'_2 - t'_1. \quad (23)$$

In this case $t'_2 - t'_1$ can be negative. This causes t'_2 to be smaller than t'_1 where the action precedes the cause.

Exercise 4: Electron-Positron annihilation

- a) In the center-of-mass frame, the two incoming particles have momenta in opposite directions along an arbitrary x -axis. The magnitude of the momenta p is equal for both. The particles also have the same energy which is denoted as E because they have the same rest mass.

Assuming there is only one outgoing photon γ , conservation of energy tells us that $E_\gamma = 2E$. Conservation of momentum, however, indicates that $p_\gamma = |\vec{p}_{\text{electron}} + \vec{p}_{\text{positron}}| = 0$. This is in contradiction with $m_\gamma = 0$, where $m_\gamma = \frac{1}{c^2} \sqrt{E_\gamma^2 - p_\gamma^2 c^2}$. In practice, two or three photons are produced to conserve energy and momentum.

- b) The total energy of the electron and of the positron is given by: $E^2 = p^2 c^2 + m_e^2 c^4$. Inserting the numbers, one finds:

$$E \approx 1.55 \text{ GeV}.$$

Thus, $E \approx pc$, as expected from $pc \gg m_e c^2$.

The momentum of the incoming particles is given by $p = \gamma m_e \beta c$ where $\beta = v/c$ and $\gamma = (1 - \beta^2)^{-1/2}$. Desired is an expression for the velocity v or β , to give it as a fraction of the vacuum speed of light:

$$\frac{p^2}{m_e^2 c^2} = \beta^2 \gamma^2 = \frac{\beta^2}{1 - \beta^2}.$$

With the definition $Q = m_e c/p$, we obtain:

$$\frac{1}{Q^2} = \frac{\beta^2}{1 - \beta^2} \Rightarrow \beta^2 = \frac{1}{1 + Q^2}.$$

Because $Q \ll 1$ (as shown above), we can develop it into a Taylor series and keep only leading terms:

$$\beta^2 \approx 1 - Q^2 \rightarrow \beta \approx 1 - \frac{Q^2}{2} = 99.999995\%,$$

i.e., the electron and the positron travel almost at the speed of light.

Finally, momentum conservation tells us that the momentum of the produced particle must be zero, and with energy conservation one obtains $2E = E_{J/\Psi} = M_{J/\Psi} c^2$. J/Ψ thus has a mass of $3.1 \text{ GeV}/c^2$.

Exercise 5: Relativistic collisions

- a) Using the momentum conservation one has :

$$\vec{p} = \vec{p}_1 + \vec{p}_2 \Rightarrow p^2 = p_1^2 + p_2^2 + 2p_1 p_2 \cos \theta \quad (24)$$

$$\Rightarrow \cos \theta = \frac{p^2 - (p_1^2 + p_2^2)}{2p_1 p_2}, \quad (25)$$

where $\vec{p}_1$ and $\vec{p}_2$ are the momenta of the particles after the collision and $\vec{p}$ is the momentum of the incoming particle.

One has now to relate this momenta to the kinetic energy and to the mass via the expression $E^2 = p^2 c^2 + m^2 c^4$. The incoming particle energy is $E = mc^2 + E_0$, while the energy for the outgoing particles is $E_1 = E_2 = mc^2 + E_0/2$. Therefore one gets:

$$p^2 c^2 = E^2 - m^2 c^4 = (mc^2 + E_0)^2 - m^2 c^4 = E_0^2 + 2E_0 mc^2, \quad (26)$$

$$p_1^2 c^2 = p_2^2 c^2 = E_1^2 - m^2 c^4 = \left(mc^2 + \frac{E_0}{2}\right)^2 - m^2 c^4 = \frac{E_0^2}{4} + E_0 mc^2. \quad (27)$$

This leads to the following expression for the angle between the outgoing particles:

$$\cos \theta = \frac{E_0^2 + 2E_0mc^2 - 2\left(\frac{E_0^2}{4} + E_0mc^2\right)}{2\left(\sqrt{\frac{E_0^2}{4} + E_0mc^2}\right)^2} \quad (28)$$

$$= \frac{\frac{E_0^2}{2}}{2\left(\frac{E_0^2}{4} + E_0mc^2\right)} \quad (29)$$

$$= \frac{E_0}{E_0 + 4mc^2}. \quad (30)$$

- b)** In the ultrarelativistic limit: $E_0 \gg mc^2$, thus $\cos \theta \rightarrow 1$, and $\theta \rightarrow 0$. On the other hand for nonrelativistic particles: $E_0 \ll mc^2$, thus $\cos \theta \rightarrow 0$, and $\theta \rightarrow \pi/2$.